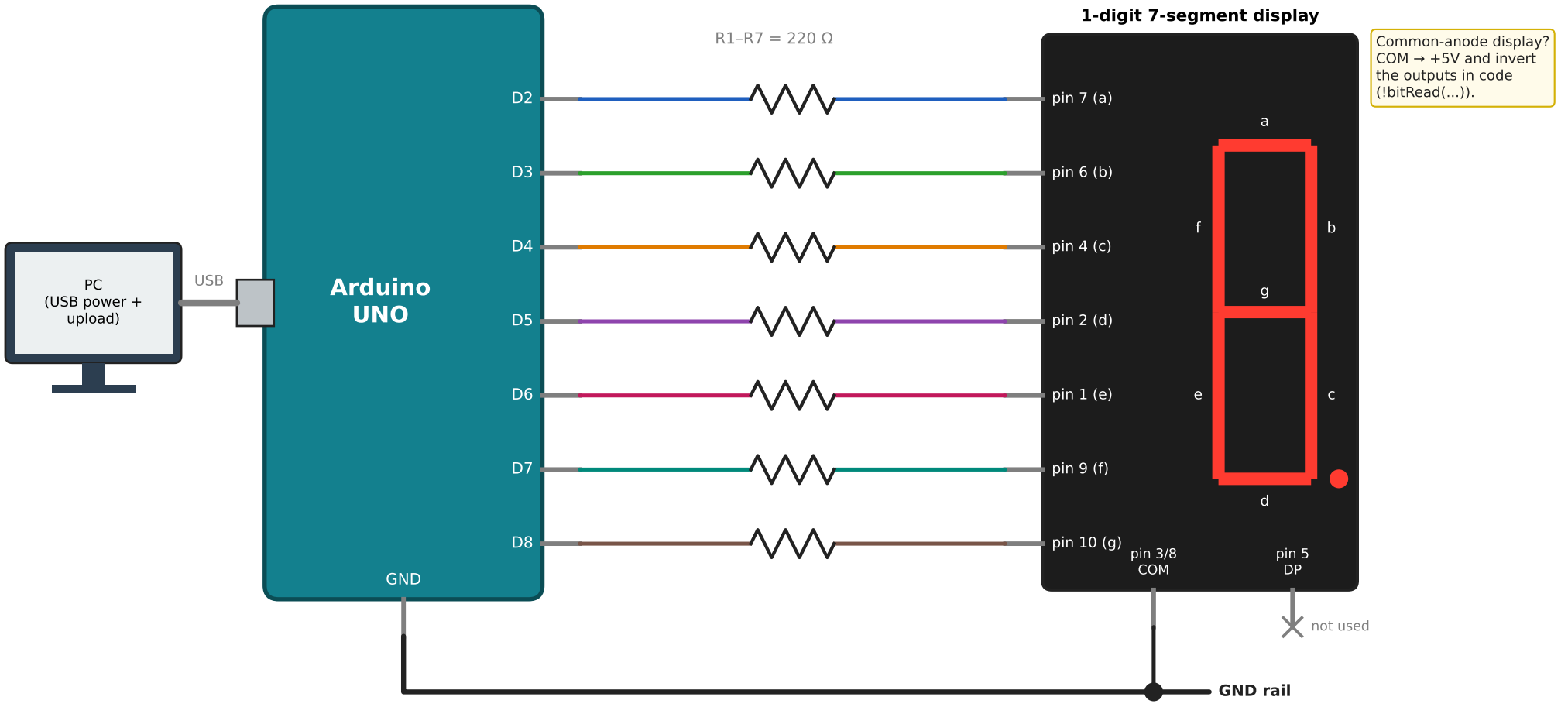
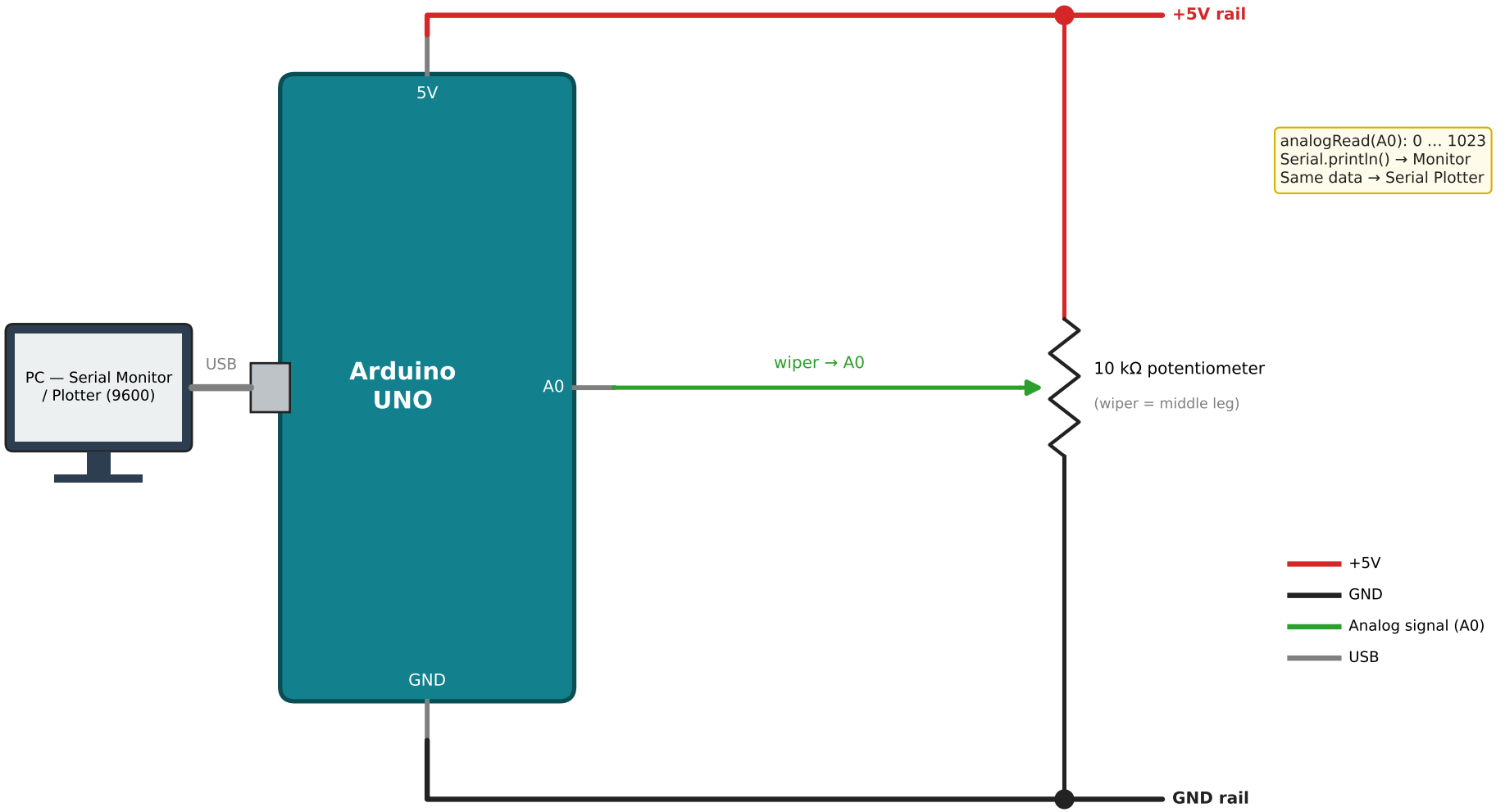
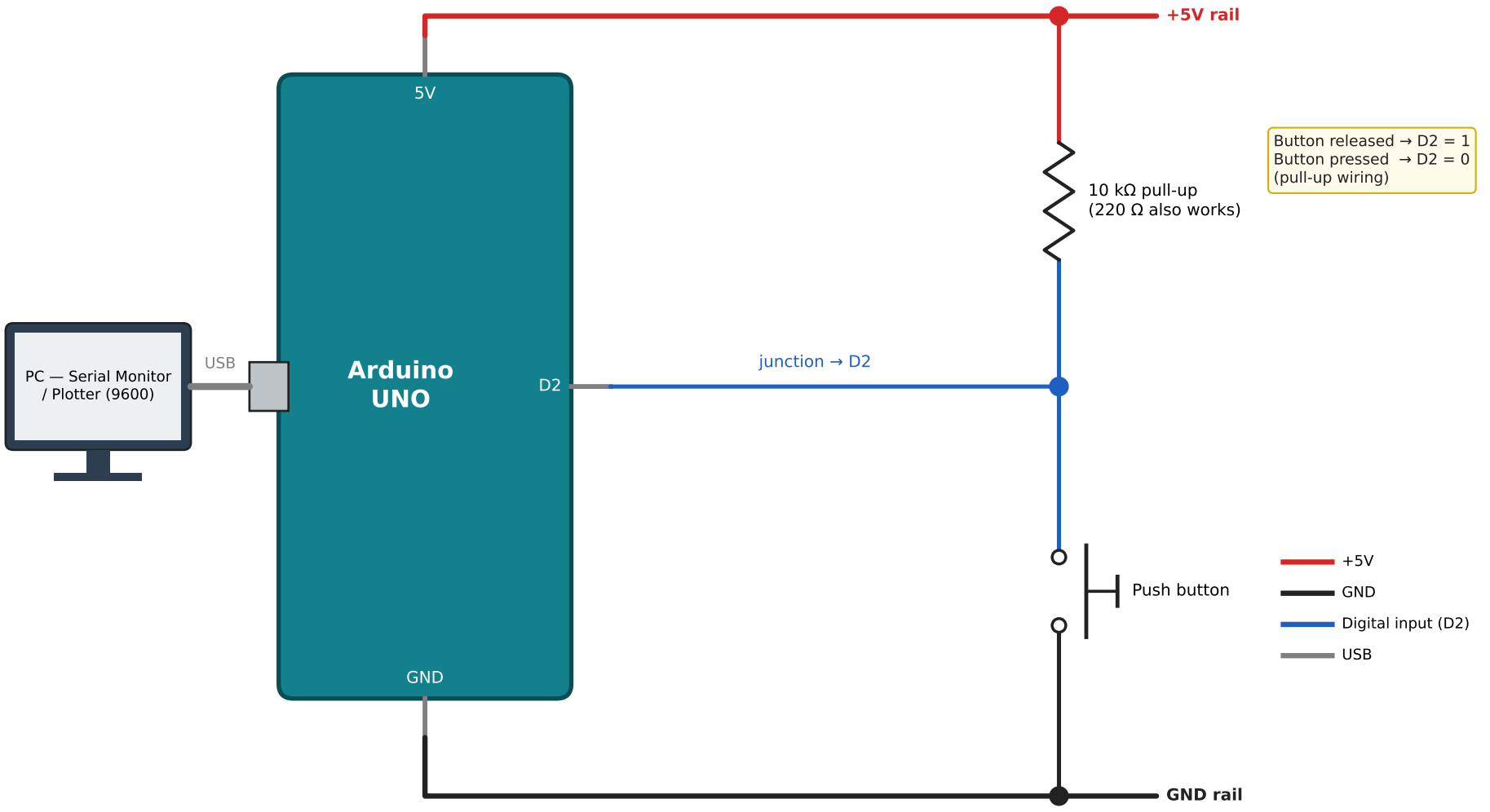
Experiment 21 — 7-Segment Display Counting 0 → 9 (common cathode)



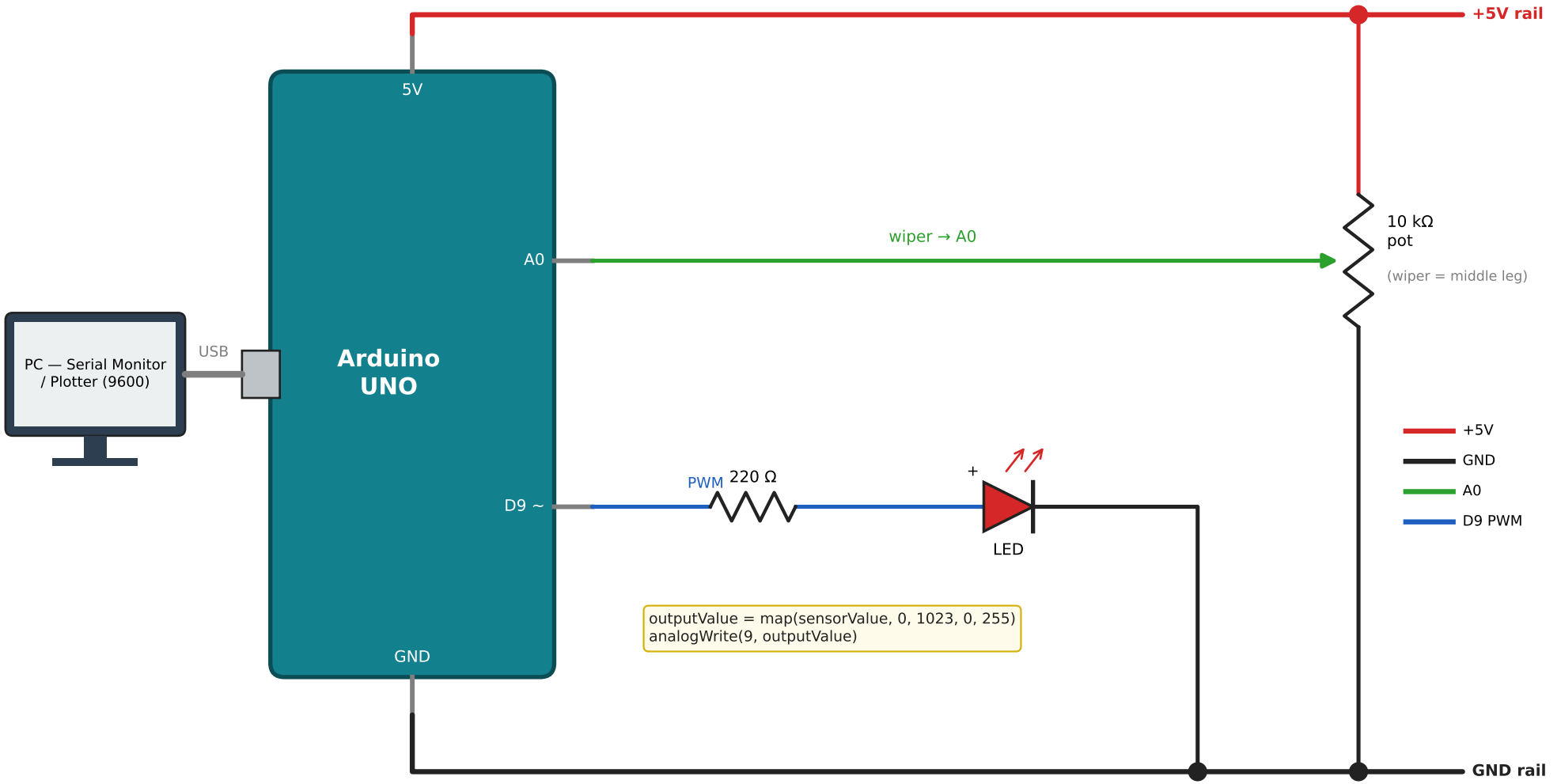
Experiment 22 — Potentiometer Value on Serial Monitor / Plotter



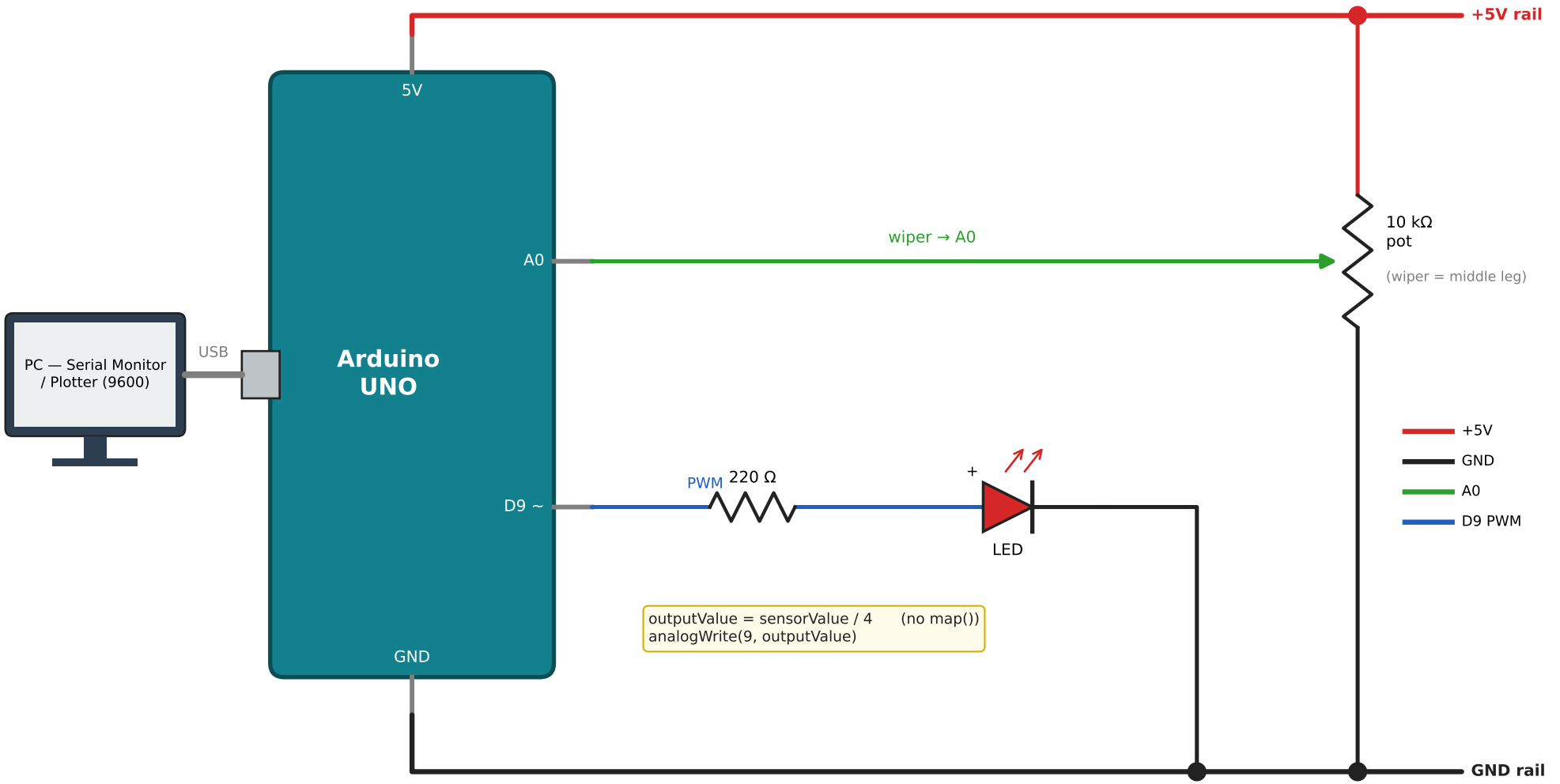
Experiment 23 — Push Button State on Serial Monitor / Plotter



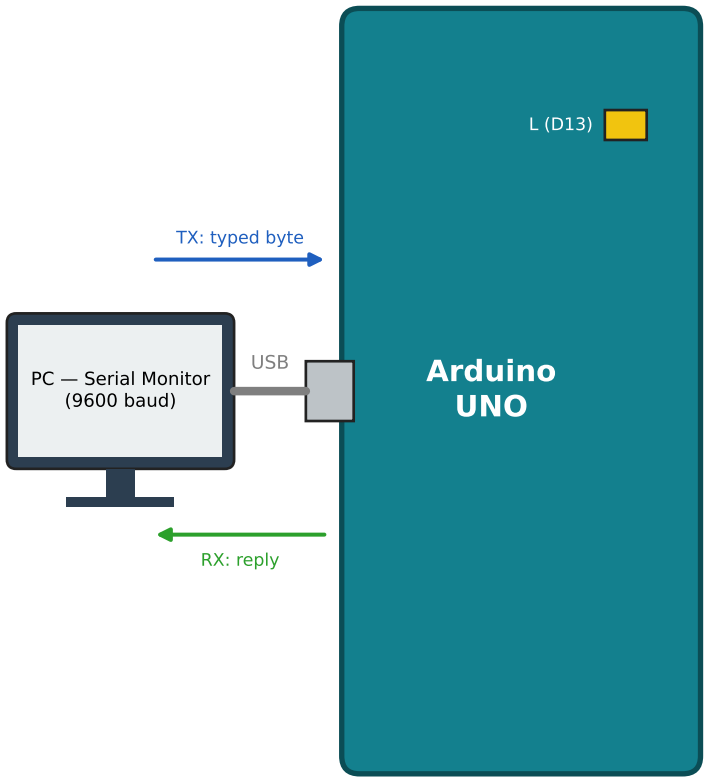
Experiment 24 — Pot → Serial Monitor + LED Brightness (map)



Experiment 25 — Same Circuit as 24, Scaling Without map()



Experiment 26 — Serial Read (typing characters to the Arduino)



Serial Monitor — COM3 · 9600 baud · Newline

A

Send

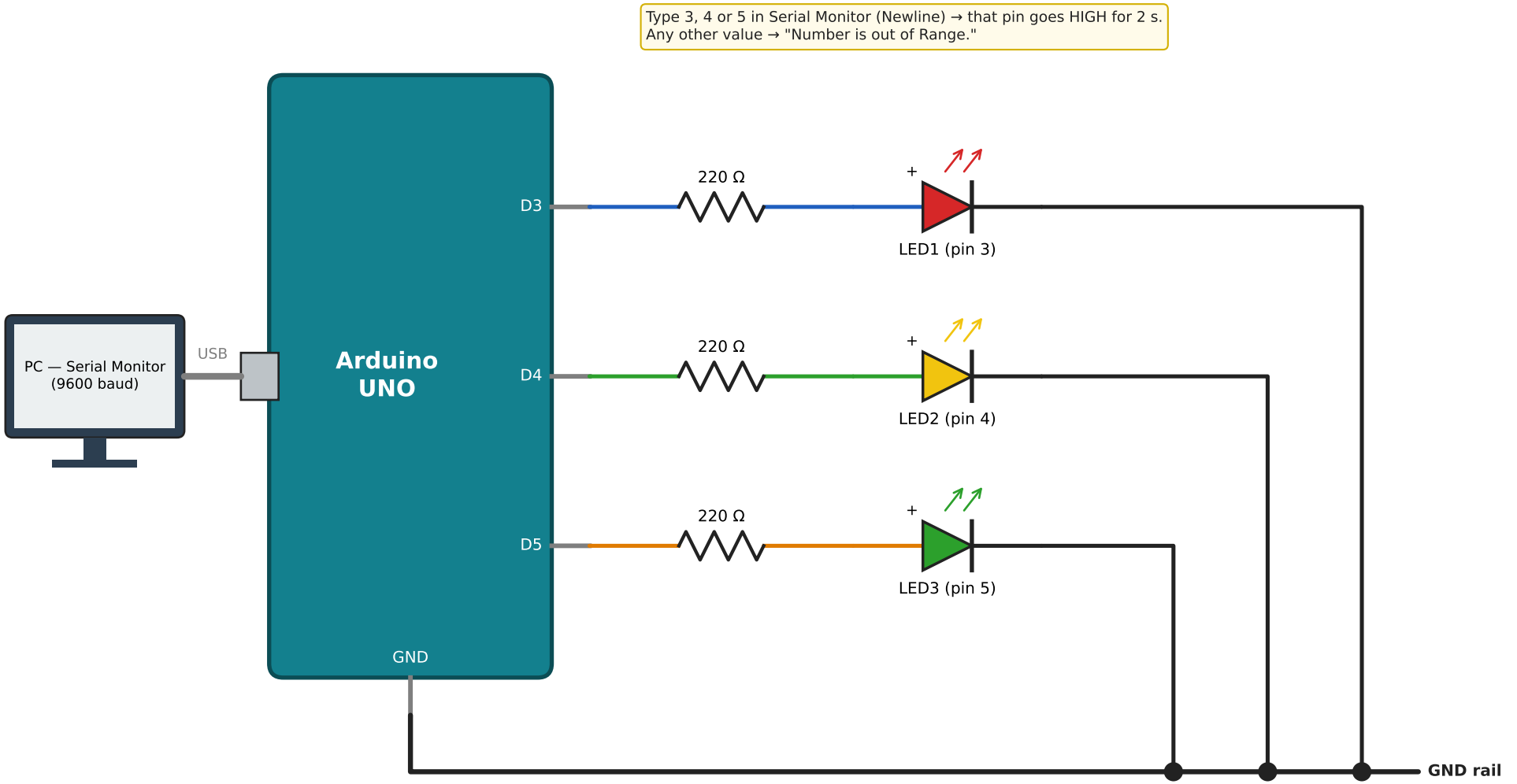
```
I received: 65   ← ASCII code of 'A'

I received: 10   ← newline '\n'

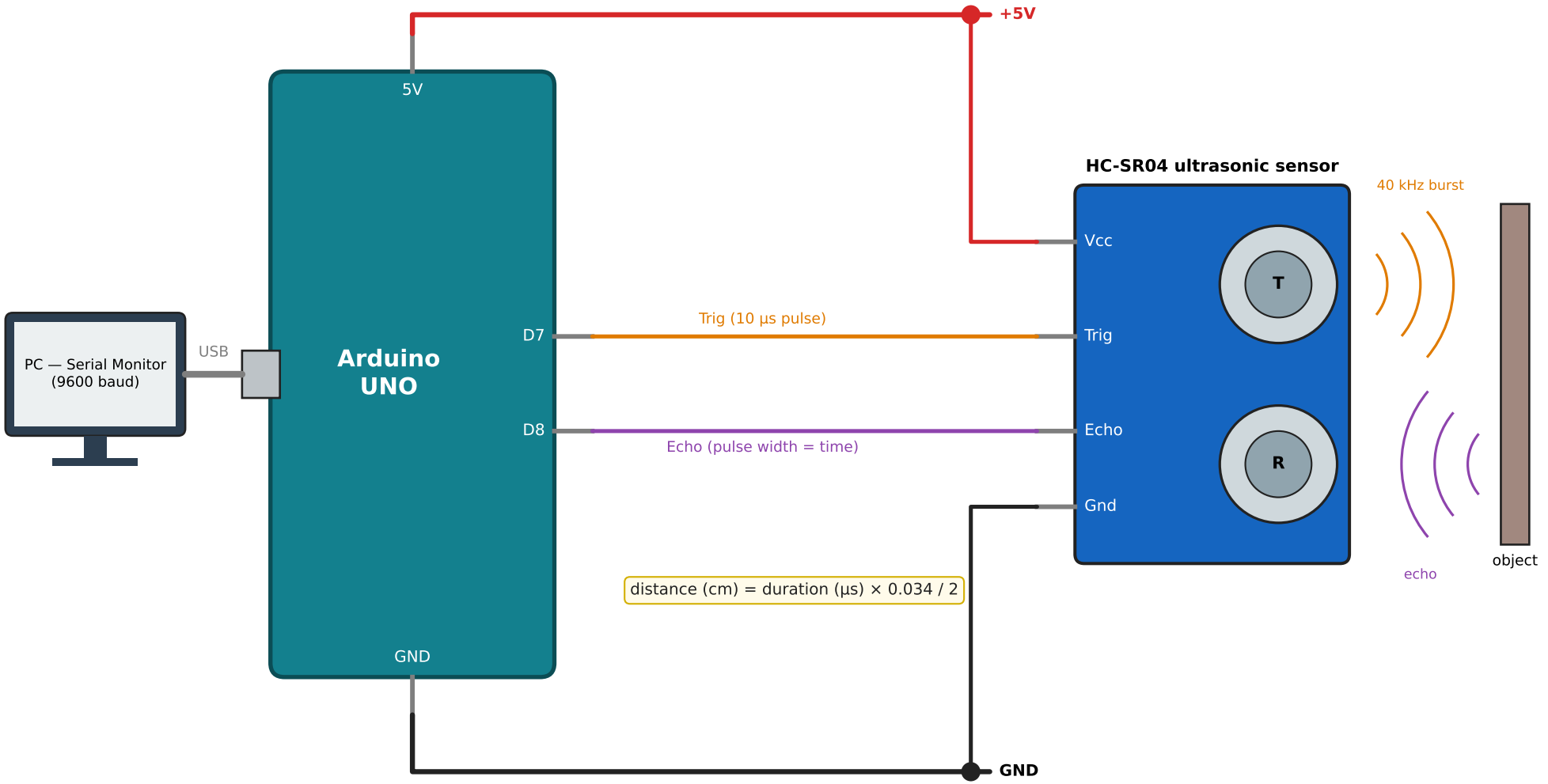
I received: 52   ← '4'
```

No external components — only the UNO on USB.
Serial.available() > 0 → Serial.read() → Serial.println(byte, DEC)

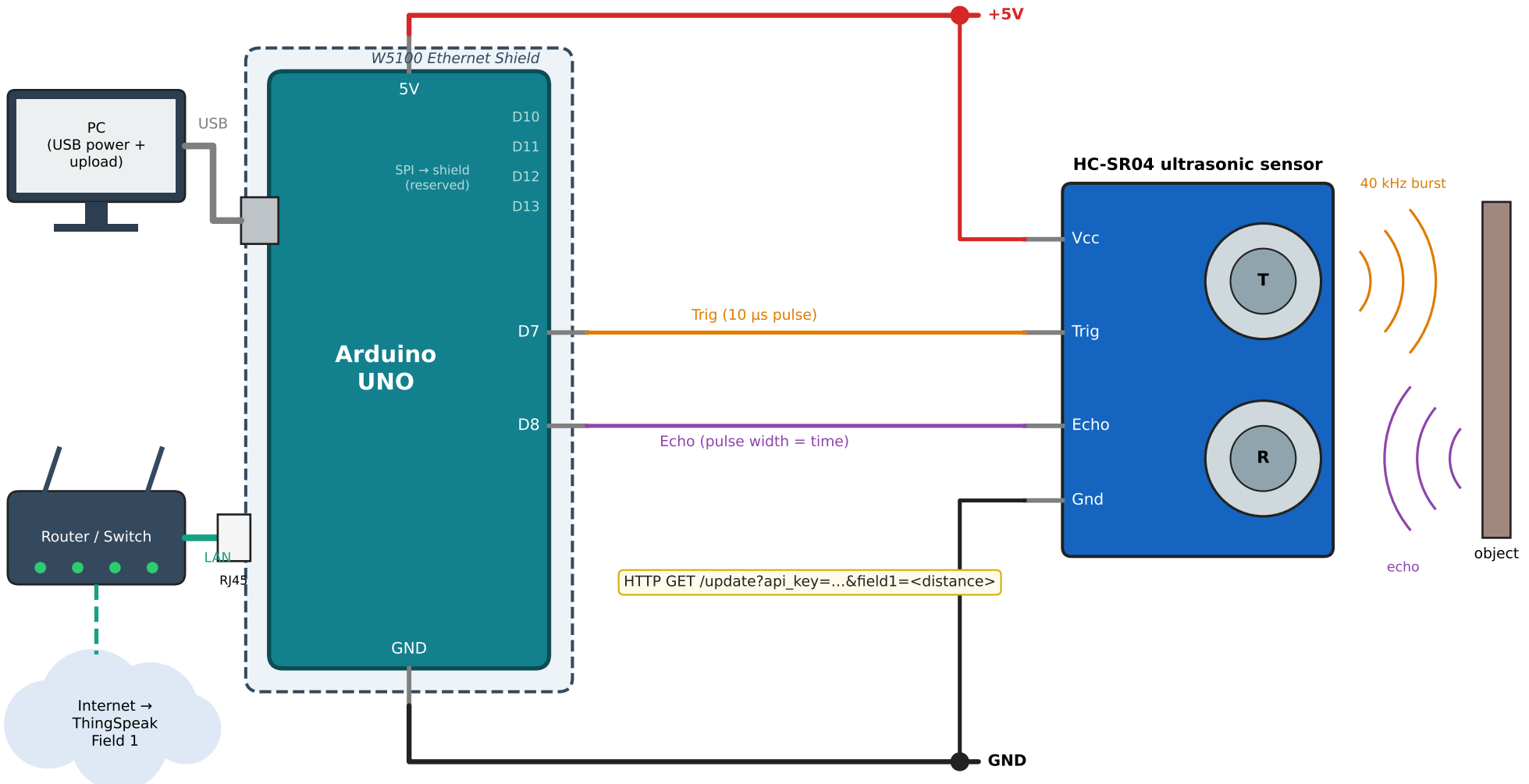
Experiment 27 — Typed Number 3 / 4 / 5 Lights That LED for 2 s



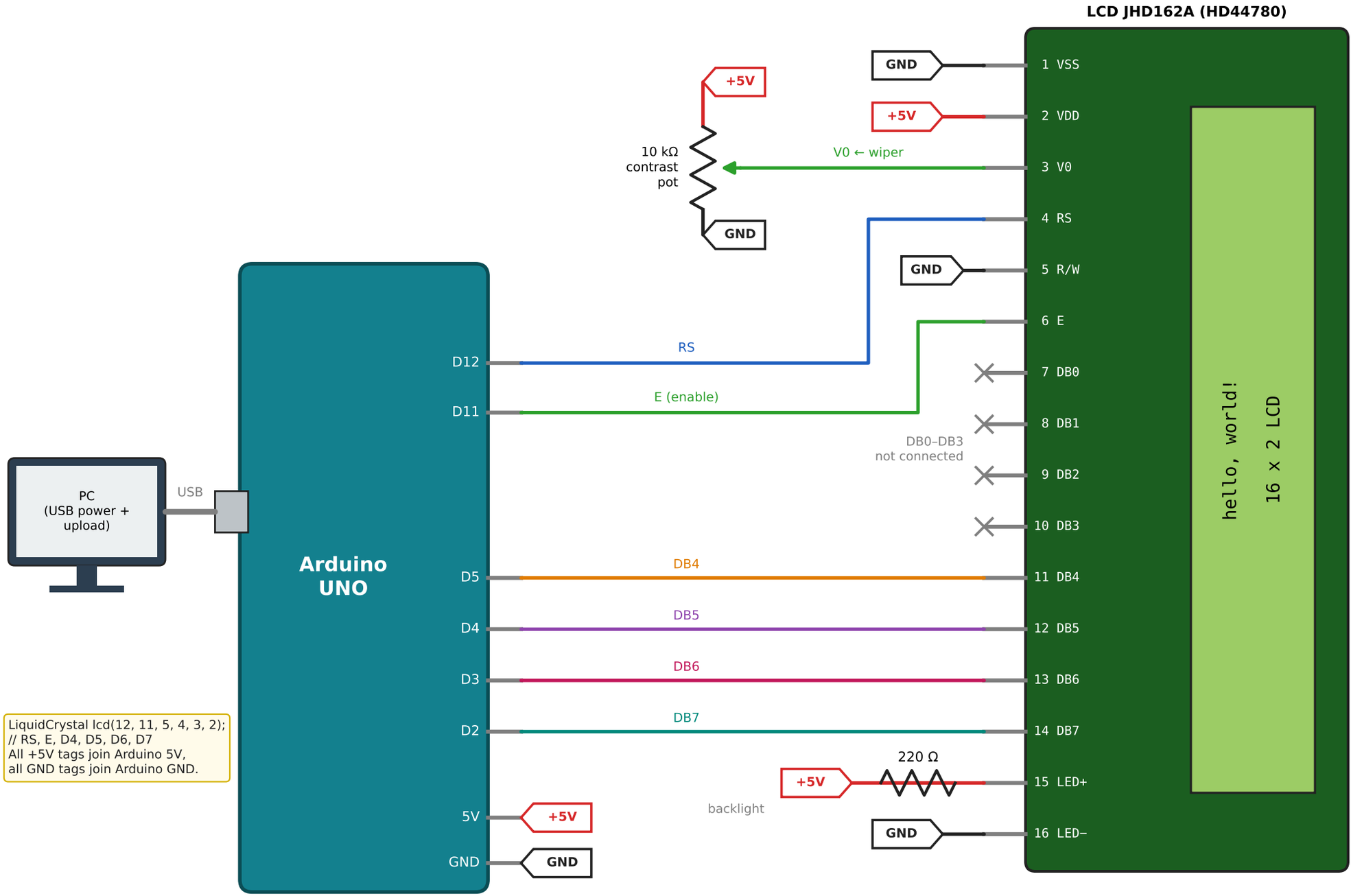
Experiment 28 — Distance Measurement with HC-SR04



Experiment 29 — Ultrasonic Distance Sent to ThingSpeak



Experiment 30 — 16x2 LCD (JHD162A) in 4-bit Mode



Experiment 31 — Arduino B Reads Arduino A's Web Page

